

Class 3 — Final Practice

You've finished the list concepts. Now let's combine them into practical problems.

□ Exercise 1 — Basic List Operations

Start with:

```
numbers = [50, 20, 40, 10, 30]
```

Do these operations in order:

1. Add 60 to the end.
2. Insert 5 at index 0.
3. Remove 40.
4. Remove the last element using `pop()`.
5. Sort the list in ascending order.
6. Reverse the list.
7. Print the final list.

Try to write the code yourself.

□ Exercise 2 — Aliasing

Predict the output **before running it**:

```
a = [1, 2, 3]
b = a
b.append(4)
print(a)
print(b)
print(a is b)
```

Think carefully about:

```
b = a
```

□ Exercise 3 — Copy

Now:

```
a = [1, 2, 3]

b = a.copy()

b.append(4)

print(a)
print(b)
print(a is b)
```

Compare this with Exercise 2.

The key difference is:

```
b = a
```

versus

```
b = a.copy()
```

□ Exercise 4 — Nested List

Given:

```
students = [
    ["Sanzid", 80],
    ["Rahim", 75],
    ["Karim", 90]
]
```

Print:

```
Sanzid
80
Rahim
75
Karim
90
```

using indexing.

For example, the first name can be accessed with:

```
students[0][0]
```

□ Exercise 5 — Unpacking

Given:

```
numbers = [10, 20, 30, 40, 50]
```

Use unpacking to get:

```
first = 10
middle = [20, 30, 40]
last = 50
```

Hint:

```
first, *middle, last = numbers
```

★ Final Challenge

Write a small program:

```
students = ["Rahim", "Karim", "Sanzid", "Nadia"]
```

Your program should:

1. Add "Amin" to the end.
2. Insert "Rafi" at index 1.
3. Remove "Karim".
4. Sort the names alphabetically.
5. Make a copy called `backup`.
6. Add "Test" to `backup`.
7. Print both lists.
8. Print whether `students` and `backup` are the same object.

The final concept you should understand is:

```
backup = students.copy()
```

means **two separate lists**, while:

```
backup = students
```

means **two names referring to the same list**.

After you finish this challenge, **Class 3 — Lists is complete**, and we can move to the next Python class.